Speaker Information

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Talk Information

Hierarchical processing and polarization encoding in the cephalopod visual system

Vertebrates and cephalopods independently evolved camera-type eyes through convergent evolution. Unlike most vertebrates, cephalopods are colorblind but detect light polarization, which aids underwater object detection and communication. However, the neural processing underlying their vision is largely unknown. We developed a novel head-fixation method to perform two-photon calcium imaging and Neuropixels electrophysiology in awake squids, providing the first in vivo recordings of the cephalopod visual system at cellular resolution. In the optic lobe cortex, we identified distinct neuron classes tuned to polarization orientation and intensity. In the optic lobe medulla, we found evidence for hierarchical processing: receptive fields expanded with depth, and intensity and polarization signals were integrated with progressively greater complexity. These findings reveal neural mechanisms of cephalopod vision and broaden our understanding of convergent evolution in visual systems.

Research Overview (for collaboration discussion)

[Research Interests] Cephalopods, Vision, Sleep

[Current Research Projects] Squid visual system

[Techniques]

Two-photon calcium imaging, Neuropixels recording, Single-cell RNA-seq, Tissue clearing and whole-brain imaging

[Aspects or techniques of interest for future collaboration] Freely moving calcium imaging

Short Bio

I obtained my B.A. in physics from the University of Tokyo, studying biophysics. I then earned a Ph.D. in Information Science & Technology from the University of Tokyo, working on tissue clearing and 3D imaging of mouse brains. In 2021, I moved to OIST as a postdoc and began working on cephalopods, an emerging model for understanding the evolution and convergence of intelligence.